

SHANKAR KUMAR

Python Developer · Django & Flask · REST API · AI/ML Integration · AWS Cloud

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PROFESSIONAL SUMMARY

Results-driven Python Developer with 2+ years of hands-on experience designing, developing, and deploying scalable web applications using Django and Flask. Demonstrated expertise in REST API development using Django REST Framework (DRF), AI/ML model integration (HuggingFace BART, GPT-4, Rasa NLU), and database optimization across MySQL, PostgreSQL, and SQL Server. Experienced in cloud infrastructure on AWS (EC2, S3), containerization with Docker, and asynchronous processing via Celery and Redis. Consistently delivers government-grade and enterprise-level software on schedule across the full SDLC — from requirements gathering and architecture design to production deployment and performance monitoring — in both independent and Agile/Scrum team environments.

CORE COMPETENCIES

Python · Django · Flask · Django REST Framework (DRF) · REST API Design · FastAPI · SQL · MySQL · PostgreSQL · SQL Server · ORM
HuggingFace BART · GPT-4 / OpenAI API · Rasa NLU · NLP · Machine Learning · AI Integration · Prompt Engineering
AWS EC2 & S3 · Docker · Celery · Redis · Git · CI/CD · OOP · Microservices · Agile/Scrum · SDLC · JavaScript · HTML5 · CSS3

TECHNICAL SKILLS

Languages: Python, SQL, JavaScript (ES6+), HTML5, CSS3, jQuery **Frameworks:** Django, Flask, Django REST Framework (DRF), BeautifulSoup
AI / ML & NLP: HuggingFace BART, GPT-4, OpenAI API, Rasa NLU, NLP, Regex Pipelines **Databases:** MySQL, PostgreSQL, SQL Server, Django ORM, Stored Procedures
Cloud & DevOps: AWS (EC2, S3), Docker, Celery, Redis, Git, CI/CD Pipelines **Architecture:** OOP, DSA, Microservices, REST API Design, SDLC, Agile/Scrum, Data Normalization

PROFESSIONAL EXPERIENCE

Python Developer · Cyfuture India Pvt. Ltd. · Greater Noida

Oct 2024 – Present

- Independently designed and developed production-grade Python web applications using Django and Flask, reducing delivery timelines by 20% through disciplined SDLC practices and modular architecture.
- Automated repetitive business workflows using Python scripting, scheduled tasks, and Celery-based async queues with Redis broker, improving team operational efficiency by 30%.
- Integrated AI/ML models including HuggingFace BART and GPT-4 (OpenAI API) into production pipelines, enabling NLP-based automation, intelligent data extraction, and conversational AI features.
- Designed and maintained RESTful APIs using Django REST Framework (DRF) with JWT authentication, rate limiting, and versioning — consumed by internal dashboards and third-party clients.
- Optimized Python codebase following OOP principles, PEP8 standards, and DRY patterns; reduced average API response time and lowered memory footprint through query optimization and caching strategies.
- Managed complete project lifecycle — requirements analysis, system design, development, testing, deployment on AWS — with consistent on-time, on-budget delivery across concurrent projects.
- Collaborated with government stakeholders to translate complex compliance requirements into scalable technical solutions meeting security, performance, and auditability standards.

Software Engineer · Chetu Inc. · Noida

Dec 2021 – Sep 2022

- Developed reusable, modular Python and Django components adopted across multiple product lines, reducing cross-project development effort by 25% and improving code maintainability.
- Built scalable backend systems with clean architecture ensuring long-term stability, ease of testing, and smooth extensibility for future feature additions.

- Integrated multiple third-party REST APIs and SDKs into existing platforms, expanding product capabilities and improving end-user workflows.
- Participated in Agile/Scrum sprints alongside QA engineers, frontend developers, and product managers, consistently meeting sprint commitments and release deadlines.
- Conducted peer code reviews, enforced coding standards and best practices, and contributed to technical documentation and internal knowledge base articles.

KEY PROJECTS

Government Grievance Automation Portal | *Python | Flask | SQL Server | HuggingFace BART | NLP | IMAP*

- Built a production-ready grievance automation platform with IMAP-based email ingestion and intelligent NLP classification pipelines, reducing manual processing effort by 80%.
- Implemented HuggingFace BART transformer model for automated email summarization, converting unstructured citizen complaints into structured, searchable SQL Server records with metadata tagging.
- Developed Flask web app with full record lifecycle management — intake, validation, assignment, status tracking, file attachments, and audit trails — with role-based access control (RBAC).
- Secured all database interactions using parameterized queries and stored procedures, preventing SQL injection and achieving significant throughput improvement over ORM-based queries.

AI Chatbot System | *Python | Flask | Rasa NLU | GPT-4 Mini | REST API | Docker*

- Developed an AI-powered domain-specific chatbot combining Rasa NLU for intent recognition with GPT-4 Mini fallback for open-ended Q&A, achieving high accuracy on domain-specific queries.
- Built dynamic document upload module enabling continuous knowledge base updates without model downtime; automated Rasa YAML config generation and retraining pipeline via Flask REST APIs.

Government Data Collection Application | *Python | Django | MySQL | REST API | JavaScript*

- Contributed to a large-scale multi-ministry data collection platform integrating secure REST APIs from 10+ state and central government departments with full encryption and data validation pipelines.
- Designed and delivered an administrator training module with role-specific workflows, adopted by 100+ users across departments, improving onboarding time by 40%.

Supply Chain Management System (SCMS) | *Django | PostgreSQL | AWS EC2 & S3 | Docker | Celery | Redis*

- Developed end-to-end SCMS covering procurement, inventory management, distribution logistics, and POS transactions; integrated Celery + Redis for async processing, boosting system throughput significantly.
- Containerized application with Docker and deployed on AWS EC2 with static assets on S3, enabling auto-scalable, environment-consistent production infrastructure with zero-downtime deployments.

Government Inventory Management System | *Python | Django | MySQL*

- Enhanced Consumable Module with automated monthly consumption limits, configurable multi-level approval workflows, real-time budget validation, and financial audit logs to prevent unauthorized spend.

EDUCATION

Master of Computer Applications (MCA)*Apr 2022* | Indira Gandhi National Open University (IGNOU), New Delhi

Bachelor of Computer Applications (BCA)*Jun 2018* | Indira Gandhi National Open University (IGNOU), New Delhi

CERTIFICATIONS & ADDITIONAL

- AWS Cloud Practitioner Essentials — Completed foundational AWS training covering EC2, S3, IAM, VPC, RDS, and cloud architecture best practices.
- Python for Data Science & AI — Completed advanced Python training covering data manipulation, REST API integration, automation, and AI/ML model deployment workflows.
- Open Source Contributor — Actively explores and contributes to Python/Django ecosystem; maintains personal projects on GitHub demonstrating full-stack and AI integration skills.
- Research & Documentation — Strong foundation in structured technical writing, research methodology, and governance/public policy; enhances solution documentation quality on enterprise and government engagements.